

The empirical recurrence for OEIS A281952, recovered from its tail

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Abstract

OEIS A281952 records “Empirical recurrence of order 69” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 874$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{69} = -16 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A281952 is “Number of nX5 0..1 arrays with no element unequal to more than four of its king-move neighbors and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

16, 420, 2559, 19089, 130416, 944967, 6763599, 48984359, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Feb 03 2017 08:23:44, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 874$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 874$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1748$ exact terms from $n = 5$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	6	c_{19}	−14415049	c_{37}	199420041	c_{55}	767910
c_2	36	c_{20}	−11516784	c_{38}	−234325839	c_{56}	−357125
c_3	−128	c_{21}	36457177	c_{39}	−64316196	c_{57}	−196324
c_4	−743	c_{22}	1002324	c_{40}	226666878	c_{58}	576550
c_5	1151	c_{23}	−57183055	c_{41}	−38200656	c_{59}	124734
c_6	6656	c_{24}	20522266	c_{42}	−162440444	c_{60}	−263715
c_7	−5777	c_{25}	49270297	c_{43}	73465288	c_{61}	−61230
c_8	−30516	c_{26}	−19079714	c_{44}	100700263	c_{62}	75088
c_9	11887	c_{27}	−14461035	c_{45}	−58788292	c_{63}	7354
c_{10}	88876	c_{28}	−25042230	c_{46}	−60097393	c_{64}	−7248
c_{11}	11003	c_{29}	11341932	c_{47}	30157213	c_{65}	−1822
c_{12}	−245663	c_{30}	63542388	c_{48}	35808123	c_{66}	1124
c_{13}	−59974	c_{31}	−83402221	c_{49}	−12504843	c_{67}	−572
c_{14}	909395	c_{32}	−49084813	c_{50}	−20245680	c_{68}	208
c_{15}	−327436	c_{33}	201412871	c_{51}	5355584	c_{69}	−16
c_{16}	−3385497	c_{34}	−28968389	c_{52}	9734826		
c_{17}	3416875	c_{35}	−263502462	c_{53}	−2773345		
c_{18}	8566864	c_{36}	148164504	c_{54}	−2295465		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A281952 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{69-1} - \dots - c_{69}$. Its constant term is $-c_{69} = 16$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -16 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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